

Report 2: GPU Device Info

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1 Goal

Use `numba.cuda` to get basic information about the GPU: its name, number of multiprocessors, number of cores and memory size.

I ran the code on Google Colab (GPU runtime) because my laptop has no NVIDIA GPU. The GPU is a **Tesla T4**.

2 Method

- **Name and ID:** `cuda.select_device(0)` gives a device object. I read `id`, `name` and `compute_capability` from it.
- **Multiprocessors:** read from `MULTIPROCESSOR_COUNT`.
- **Cores:** Numba does not give this number directly. Cores per SM depend on the compute capability, so I used a small lookup table:

$$\text{total cores} = \text{number of SMs} \times \text{cores per SM}$$

- **Memory:** `cuda.current_context().get_memory_info()` returns free and total memory in bytes.

3 Code

```
from numba import cuda

# compute capability -> cores per SM
CORES_PER_SM = {(6, 0): 64, (6, 1): 128, (7, 0): 64,
                 (7, 5): 64, (8, 0): 64, (8, 6): 128}

device = cuda.select_device(0)
print("ID:", device.id)
print("Name:", device.name)

cc = tuple(device.compute_capability)
sm = device.MULTIPROCESSOR_COUNT
print("SM count:", sm)
print("Total cores:", sm * CORES_PER_SM[cc])

free, total = cuda.current_context().get_memory_info()
print("Total memory (MB):", total / 1024**2)
print("Free memory (MB):", free / 1024**2)
```

4 Results

Device name	Tesla T4
Compute capability	7.5
Multiprocessors	40
Cores per SM	64
Total cores	2560
Total memory	14912.69 MB ($\approx$ 14.56 GB)
Free memory	14807.81 MB

5 Conclusion

The Tesla T4 has 40 SMs with 64 cores each, so 2560 cores in total, and about 14.56 GB of memory. The number of cores is not given directly by Numba; it must be calculated from the compute capability.